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Honor Pledge

I, **Reema Sankar Ravisankar Thulasi**, **CB.EN.U4ECE23262**, do hereby declare that I have neither taken nor provided any help, from any source, for the purpose of completing this assignment. I understand that typing my full name in the space below, is equivalent to affixing my signature to this statement and that it is binding.

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CLOUD-ASSISTED ADAPTIVE LoRa COMMUNICATION SYSTEM FOR REMOTE HEALTH MONITORING

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***Abstract*—Remote health monitoring over LoRa spans four layers: physiological sensing, clinical inference, adaptive link control, and delivery to a cloud record. Reviews take one layer at a time. We read all four across 24 DOI-verified works published between 2017 and 2026, compared on five structural axes. Link adaptation is driven almost entirely by radio observables: exactly one of the 24 works lets a patient-derived variable select a physical-layer parameter, and six of the seven deployed end-to-end systems transmit on a configuration fixed at design time. The field needs a stated policy mapping clinical urgency onto all four LoRa parameters.**

***Index Terms*—LoRa, LoRaWAN, low-power wide-area networks, Internet of Medical Things, remote health monitoring, adaptive data rate, spreading factor, duty cycle, literature review.**

I. Introduction

A patient monitored at home, in a rural clinic or in transit is only monitored to the extent that the data arrives. Everything upstream of that — the sensor, the filter, the classifier — is contingent on a link that has to cross kilometres, run from a battery, and survive walls, bodies and weather. This is the problem low-power wide-area networking was built for [1], and among the available technologies LoRa is the one most often chosen for it, because chirp spread spectrum buys range and receiver sensitivity at a data rate that physiological summaries can live within [2], [3].

What makes LoRa interesting also makes it awkward. Its range, airtime, energy cost and robustness are not properties of the technology but of four parameters chosen per transmission: spreading factor, bandwidth, coding rate and transmission power. Raising the spreading factor improves receiver sensitivity and extends range while lengthening time-on-air, which costs energy and consumes a regulated airtime allowance shared with every other node in range [4], [5]. No setting is right in general: a configuration correct for a patient two kilometres from the gateway wastes airtime for one in the next room, and one tuned for the next room leaves no margin at two kilometres.

A substantial literature has therefore grown around choosing those parameters automatically, and a second, largely separate literature has grown around building LoRa health-monitoring systems and reporting what they achieve. Both are productive. The question this review asks is what happens at their boundary, and the answer is that in almost every case nothing does. The adaptation literature optimises against radio observables — signal-to-noise ratio, packet error rate, collision state, network load — and the system literature transmits on a configuration fixed at design time. The clinical significance of the payload, which is the one quantity that should determine how much airtime and energy a packet is worth, reaches the communication decision in one of the 24 works surveyed here.

We read five classes of work side by side over 24 DOI-verified works published between 2017 and 2026: physical-layer parameter adaptation, network-level resource allocation, deployed end-to-end systems, mobility and tracking, and the physiological inference layer itself. Section II states the verification protocol, the taxonomy of Fig. 1 and the five comparison axes; Section III surveys each class; Section IV sets the works against one another; Section V states the resulting open challenges and this review's limitations.

The contribution is threefold: a taxonomy spanning the sensing, adaptation and delivery layers rather than one of them, which is what makes the boundary between them visible; a comparison framework whose axes are structural rather than numerical, because the five classes report in metrics that provably do not convert; and a set of countable claims, each checkable against Table II rather than offered as an impression.

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II. Review Methodology and Taxonomy

A. Research Questions

Three questions shaped this review. RQ1 asks which quantities the literature actually adapts in a LoRa health-monitoring link, and what evidence each adaptation decision is driven by. RQ2 asks whether the clinical state of the patient ever reaches that decision, or whether link control is settled purely on radio observables. RQ3 asks what prevents the sensing, adaptation and delivery stages from being evaluated as one system.

B. Identification and Verification of Sources

Candidates were assembled in two passes. The first queried the Crossref REST API for work on the sub-problems the field is active on: spreading-factor and data-rate selection, transmission-power control, scheduling and collision avoidance, gateway placement, wearable physiological sensing and respiratory or cardiac parameter estimation. The second pass was manual, adding the survey and measurement papers that document the instruments against which the admitted works report, so that every protocol named here is sourced to a published record rather than to secondary description.

Every candidate was then checked individually. Its DOI was resolved against the Crossref works endpoint, and authors, title, container title, volume, issue and year were copied from the returned record rather than from a secondary source. A candidate whose DOI failed to resolve, or resolved to a different work, was discarded rather than corrected: two otherwise relevant LoRa healthcare papers were removed at exactly this step, because their publishing journals do not deposit with Crossref, and no claim here rests on them. All 24 admitted works carry a resolvable DOI, as do the six background references cited for context.

C. Inclusion and Exclusion Criteria

Table I states the protocol in full. Two decisions in it deserve to be pulled out, because both depart from the five-year, journals-only convention.

The publication window is 2017–2026. This is a property of the subject, not a relaxation of standards: LoRaWAN was first

specified in 2015, the measurement papers that established what the technology can and cannot carry appeared in 2016–2017 [1], [2], [4], [5], and the first healthcare deployments follow immediately after [6], [7]. Truncating to 2021 onward would have removed the foundational deployment work and left a review of adaptation algorithms with no deployment literature to compare them against, which is precisely the comparison this paper exists to make.

Conference papers were admitted alongside journal articles, giving 17 journal articles and 7 conference papers. This too follows the subject: the end-to-end system implementations, the part of the literature that reports what a real patient-side node achieves over a real link, are published predominantly at conferences, and a journals-only protocol would have selected almost exclusively for simulation studies. Admitting both is what makes the asymmetry reported in Section IV visible at all.

D. Taxonomy

The admitted works divide into the five classes of Fig. 1, cut by the layer of the monitoring chain each work intervenes at. Class A adapts physical-layer radio parameters: spreading factor (SF), bandwidth (BW), coding rate (CR) and transmission power. Class B leaves those parameters alone and intervenes above them, in scheduling, medium access, relaying or gateway placement. Class C builds complete patient-to-dashboard systems and reports what they achieve as deployed artefacts. Class D treats mobility, localisation and patient tracking, where position rather than a physiological reading is the payload. Class E is the physiological inference layer itself, estimating a clinical parameter from a sensor waveform independently of any radio.

E. Comparison Framework

A review confined to one of those classes could rank its members by a single number, because within a class the reporting protocol is effectively fixed. Across five classes no such number exists. Classes A and B report packet delivery ratio, airtime, energy per packet, age of information and queueing delay, mostly from simulation; class C reports received signal strength, delay and achieved range from hardware trials; class E reports estimation error in breaths per minute against an annotated clinical recording. None of these scales converts into another, and tabulating them side by side would imply an ordering the measurements cannot support.

The comparison in Section IV is therefore structural, on five axes. *Adaptation variable* names the quantity a work actually changes at run time, or records that nothing is changed. *Adaptation signal* names the observable that drives that change, which is the axis on which RQ2 turns. *Evidence base* separates analytical models, network simulation, hardware testbeds and public clinical datasets, because a result obtained in one of those does not transfer to another unexamined. *Reported metric* belongs to the class rather than to the individual work. *Clinical coupling* records whether any patient-derived variable reaches the communication decision, which is the question the rest of this review turns on.

III. Survey of Approaches

A. Composition of the Corpus

Applying the protocol of Table I gave the corpus Fig. 2 summarises. The year histogram has the shape the window was chosen to admit: a thin but essential tail of three works in 2017, rising to five in 2024 and six in 2025. Two thirds of the corpus (17 of 24) is journal-published; the remaining seven are conference papers, of which six sit in the system-implementation and tracking classes, which is the asymmetry Section IV returns to. The second panel splits the works across the five classes: six on physical-layer adaptation, seven on network-level resource allocation, seven end-to-end systems, and two each on tracking and on physiological inference.

TABLE I
Review Protocol: Inclusion and Exclusion Criteria

Criterion	Specification
Layers in scope	Physical-layer LoRa parameter adaptation; network-level scheduling, access and topology; end-to-end LoRa healthcare systems; mobility and patient tracking; physiological inference from sensor waveforms
Excluded	Non-LoRa LPWAN proposals evaluated without a LoRa baseline; hospital-interior short-range monitoring; clinical-outcome and health-economics studies
Publication window	2017–2026, set by the subject: LoRaWAN capability measurements and the first healthcare deployments both date from 2016–2017
Publication type	Peer-reviewed journal articles and conference papers; no preprints or book chapters. The system-implementation literature is predominantly conference-published, so a journals-only rule would have selected only for simulation studies
Verification	DOI resolved against the Crossref REST API; every bibliographic field copied from the returned record
Discarded	A candidate whose DOI failed to resolve was discarded rather than corrected; two LoRa healthcare papers were removed on this ground
Outcome	24 works admitted: 17 journal articles and 7 conference papers, 2017–2026, every entry carrying a resolvable DOI; 6 further DOI-verified works cited as background only

B. Class A: Physical-Layer Parameter Adaptation

This class treats the LoRa parameter set as a control surface. Augustin *et al.* [2] give the mechanics the class works against: SF sets symbol duration and therefore both receiver sensitivity and time-on-air, BW trades sensitivity against nominal data rate, and CR buys error resilience with redundancy.

Chache *et al.* [8] isolate CR and ask what it does to age of information, combining an analytical model with off-the-shelf transceiver measurements, and find that an appropriately chosen rate minimises age of information at a given SNR rather than monotonically helping or hurting. González-Palacio *et al.* [9] observe that the standard adaptive data rate (ADR) mechanism, surveyed in [10], keys off the maximum of past SNR values and so degrades exactly where SNR is volatile; their scheme predicts current SNR from environmental covariates instead, trained on a six-month in-field campaign, and reports a 47.1% energy reduction at 99% packet delivery ratio. Kufakunesu *et al.* [11] attack the same baseline from the interference side, allocating SF randomly to decorrelate collisions, and Leonardi *et al.* [12] replace the ADR decision rule with a long short-term memory network over the link history. Fernandes *et al.* [13] adapt one level up, switching medium-access mode under load, on the argument that the ALOHA rationale underlying most LoRa access performs poorly at scale.

Taleb *et al.* [14] are the exception that defines this review's central finding. Their SF selection for LoRaWAN-based wireless body area networks takes distance, packet error rate and energy as inputs, as the rest of the class does, but it also takes the patient's Early Warning Score, and it is evaluated on real recordings drawn from the MIMIC database. A more critical patient is assigned a more robust configuration, paying energy for reliability where reliability is clinically worth more. No other work in the corpus lets a clinical variable select a physical-layer parameter.

C. Class B: Network-Level Resource Allocation

These seven works leave SF, BW and CR to the ADR mechanism and intervene in how transmissions are ordered, relayed or collected. Ali *et al.* [15] cluster traffic with a Gaussian mixture model and schedule against the result, reporting reduced collision rate and delay with improved packet success ratio over conventional LoRaWAN scheduling. Kang *et al.* [16] note that the star topology loads edge-of-cell nodes most heavily and insert relays, trading range for lifetime. Mahalakshmi and Lalithamani [17] treat gateway siting as the decision variable for an Internet of Medical Things

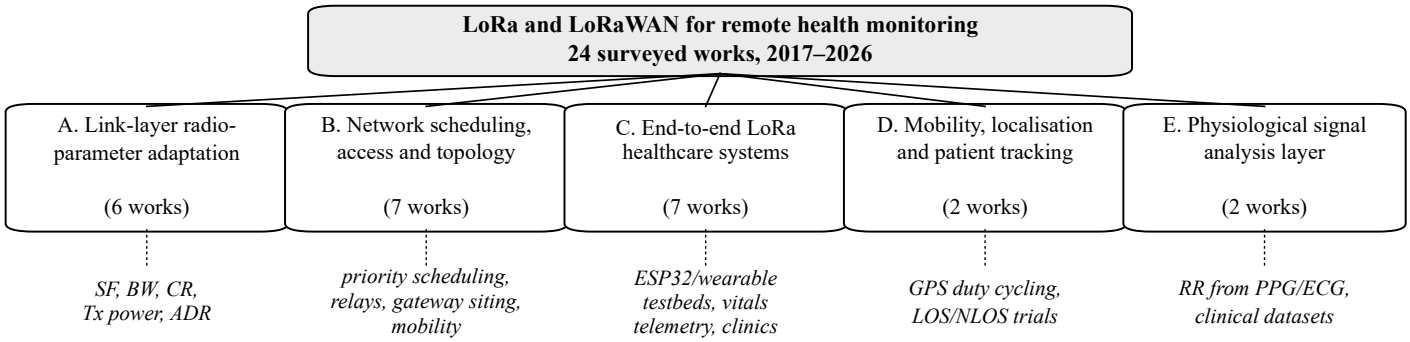


Fig. 1. Taxonomy of the reviewed literature. The five classes are cut by the layer of the monitoring chain at which each work intervenes, from physical-layer parameter control (A) through network-level resource allocation (B) and deployed end-to-end systems (C) to mobility and tracking (D) and the physiological inference layer (E). Counts are over the 24 surveyed works.

(IoMT) network, optimising coverage, connectivity and energy in OMNeT++ with FLoRa. Sarma Lanka and Kothandaraman [18] target the mobility case that conventional LoRaWAN handles poorly, which matters directly for ambulatory patients even though their evaluation is industrial.

Three members of this class do couple to patient state, and all three do it at the scheduling layer rather than at the radio. Dewa *et al.* [19] schedule a dense healthcare IoT network by message passing under a max-product rule, maximising a priority metric and throughput together without a central coordinator, on the premise that data bearing on heart failure or fall risk must go first. Ahammed and Manjaiah [20] combine a priority scheduler for body-area networks with modified-shift-rows encryption and an authentication check, transmitting only when the signal deviates from its established level; this is the only work in the corpus that treats confidentiality of the payload at all. Anand and Karthikeyan [21] place dynamic-priority task scheduling and adaptive resource allocation at the edge, on the argument that cloud round-trips are unsuitable for latency-bound healthcare tasks.

D. Class C: End-to-End Healthcare Systems

This class reports what deployed artefacts achieve. Four works establish and then consolidate the basic architecture, a sensing node reporting through a gateway to a dashboard: Mdhaffar *et al.* [6] on feasibility, Misran *et al.* [22] carrying temperature, ECG, oxygen saturation and pulse rate from a commercial kit, Bandyopadhyay *et al.* [23] body temperature and pulse rate, and Jawad *et al.* [24] a fuller design account of the same pattern. Abdulmalek *et al.* [25] are the most thoroughly evaluated: a hybrid internet-of-wearable-things system carrying blood pressure, heart rate and body temperature over both Wi-Fi and LoRaWAN, assessed on communication reliability, latency, power consumption and sensor accuracy.

Two works in this class extend past pure telemetry. Patel *et al.* [26] add fall detection and emergency alerting to vital-sign monitoring and evaluate against public hospital datasets, so a clinical inference does drive an alert, though never a transmission parameter. Sayor and Rahman [27] address rural community clinics with an ESP32 node carrying AD8232 ECG, MAX30102 pulse-oximetry and DS18B20 temperature sensing, using Wi-Fi where it exists and falling back to LoRa where it does not. That fallback is the only adaptive behaviour anywhere in this class, and it is a binary choice of bearer rather than a parameter decision. The remaining six systems transmit on a fixed configuration.

E. Class D: Mobility, Localisation and Tracking

Both works here carry position rather than physiology, and both are energy-bound. Nugraha *et al.* [28] trial a multi-node LoRa system for tracking patients with mental disorders across line-of-sight, non-line-of-sight and crowded environments, and report communication losses at several measured locations: a useful negative result, being one of the few places in the corpus where a LoRa healthcare link is shown failing in the field. Hadwen *et al.* [7] profile component-level energy in a dementia tracker, noting that the best contemporary device managed under ten hours on a large battery, and duty-cycle the GPS receiver against LoRa reporting to extend it. Their adaptation is of the sensing schedule, not the radio configuration.

F. Class E: The Physiological Inference Layer

Two works estimate a clinical parameter from a waveform with no radio in the loop, and they matter here because they set the standard of evidence the communication literature does not meet. Pimentel *et al.* [29] estimate respiratory rate from photoplethysmography, and their contribution is precisely about robustness: distinguishing high- from low-quality input and validating on independent data, using CapnoBase and MIMIC-II. Lin *et al.* [30] fuse respiratory-modulated components from ECG and PPG with a principal-component method and respiratory-quality assessment, evaluated on CapnoBase and BIDMC with 42 and 53 subjects respectively. Both report error against annotated ground truth on named public datasets with stated subject counts. No work in classes A–D reports its clinical layer to that standard.

IV. Comparative Analysis

Table II places all 24 works on the five axes of Section II. Read by column, three patterns emerge that no single class exhibits on its own.

A. Adaptation Is Driven by the Radio, Not by the Patient

Read down the adaptation-signal column and the observables are almost uniformly radio-side: SNR, received power, packet error rate, collision state, queue occupancy, network load. Of the 24 works, exactly one lets a patient-derived variable select a physical-layer parameter: Taleb *et al.* [14], where the Early Warning Score participates in SF selection. Three works couple clinical state to a communication decision at all, the other two being Dewa *et al.* [19] and Ahammed and Manjaiah [20], and both act on transmission order rather than on link configuration.

That ratio needs its denominator stated honestly. Eleven of the 24 works adapt nothing, and two of the adaptive ones [13], [16] target industrial or generic LPWAN deployments where coupling to a patient would be out of scope; counted over the 13 works that adapt anything at run time, the figure is one in thirteen. Either way the conclusion holds: the field has built capable link controllers and capable physiological classifiers, and has almost never wired the second into the first. A system that knows it is carrying a

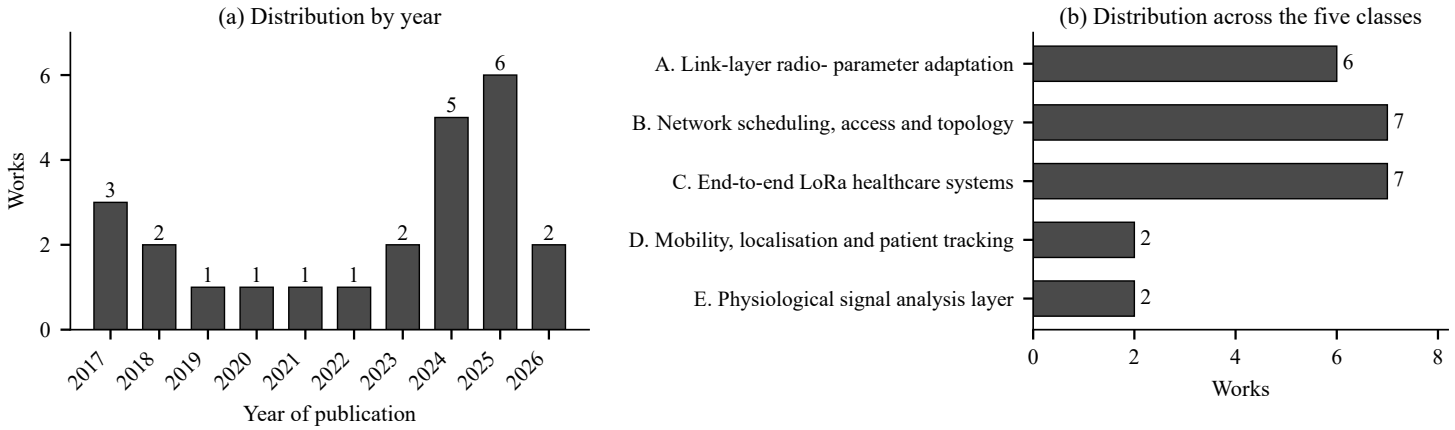


Fig. 2. Composition of the reviewed corpus, counted from the verified bibliographic record of each admitted work. (a) Distribution by publication year; every entry falls inside the 2017–2026 window on which the current-state claims rest. (b) Distribution across the five classes of Fig. 1.

tachycardia event and still picks its spreading factor from received power alone is leaving unused the one quantity that should decide how hard to work for the packet.

B. The Two Literatures Do Not Share an Evidence Base

The evidence-base column separates the corpus almost perfectly along the class boundary. Classes A and B are predominantly simulation: FLoRa under OMNeT++, custom network simulators, analytical models, with González-Palacio *et al.* [9] and Chache *et al.* [8] the partial exceptions that carry field or bench measurement. Classes C and D are almost entirely hardware: ESP32-class nodes, commercial sensor front ends, single gateways, campaigns measured in metres and hours. Class E is neither, reporting against annotated public clinical recordings.

The consequence is that the works proposing adaptation and the works demonstrating deployment have no measurement in common. An ADR variant reporting a 47.1% energy saving in simulation [9] and a clinic system reporting RSSI against distance [27] cannot be composed, because neither reports a quantity the other also reports. Not one work in the corpus reports an end-to-end figure for the chain from sensor waveform to cloud record, and so the way error and delay accumulate across that chain is simply unmeasured.

C. Scale Asymmetry Between Proposal and Demonstration

The two literatures also disagree about how large the problem is. The network-level works are evaluated at hundreds to thousands of nodes, because collisions, ADR instability and gateway siting only become interesting at that density [11], [15], [17], [19]; the deployed healthcare systems are one gateway and a handful of nodes [22]–[24], [27]. Together this means the regime in which the scheduling proposals matter has never carried real patient traffic, and the regime that has carried it is too sparse for those proposals to change anything. The duty-cycle ceiling [4] and the scaling limits [5] bind in the first regime and are invisible in the second.

D. Implications for Integrated Systems

Read as one chain, the corpus implies an ordering of design effort. The adaptation decision belongs on the patient-side node, after classification and before transmission, because that is the only point where the clinical label and the link estimate are both available; Taleb *et al.* [14] is the existence proof and the edge placement argued in [21] the architectural counterpart. Airtime, not energy, is the binding constraint once a duty cycle is enforced [4], which makes SF the parameter of largest consequence. Classification quality then floors the whole scheme, since a configuration chosen from a misclassified state is worse than a fixed one, which is where the validation discipline of class E [29], [30] has to be imported rather than admired. And bearer fallback [27] and mobility handling [18] belong in one controller, since for an ambulatory patient the choice of bearer and of spreading factor is one decision against one link estimate.

V. Open Challenges and Future Directions

The patterns of Section IV point to four problems that no single class can resolve on its own. Table III restates them against the evidence in this corpus and records which classes each affects.

1) No end-to-end evaluation protocol: Packet delivery ratio, airtime, energy per packet, age of information, queueing delay, RSSI against distance and respiratory-rate error are each well chosen for the class reporting them, and no two compose. Nothing in this corpus scores a chain from sensor waveform through classification and adaptive transmission to cloud record, so the way a classification error interacts with a configuration choice, or a duty-cycle deferral with a deteriorating patient, is unmeasured. This challenge matters most, being the one that prevents integrated systems from being compared at all. What such a protocol needs is not exotic: a clinically labelled recording, a reproducible channel condition, and one figure of merit spanning classification correctness and delivery latency.

2) Clinical state is not an input to link control: This is not technically hard: a classifier already runs on the patient-side node in several reviewed systems [26], [27] and its output is available before the transmission decision. What is missing is a stated policy mapping clinical urgency onto a configuration, and an evaluation showing the mapping is worth its cost in airtime. Taleb *et al.* [14] demonstrate it for spreading factor; nothing generalises it to the multi-parameter case where SF, BW, CR and power are chosen together.

3) The duty cycle versus clinical latency: Regulatory airtime limits in the unlicensed sub-GHz bands bound how often a node may transmit, independently of how urgent its payload is [4], [5]. For periodic telemetry that is a throughput question; for a monitoring system it is a safety one, since a node that has exhausted its allowance cannot report a deterioration it has just detected. No corpus work reports the latency distribution of an urgent event under an enforced duty cycle, and the reliance on mean energy and mean delivery ratio [9], [11], [12] obscures exactly the tail behaviour that would matter clinically.

4) Scale and security asymmetries: The scheduling and placement proposals are evaluated at densities the healthcare deployments never reach, and those deployments under conditions where the proposals cannot show an effect [15], [17], [19]. Closing this needs either patient traffic models dense enough to exercise the network proposals or deployments larger than a single gateway; neither exists here. Security is thinner still: one work in 24 treats confidentiality and authentication [20], despite every class C system carrying identifiable physiological data over a shared unlicensed

TABLE II
THE TWENTY-FOUR SURVEYED WORKS ON FIVE STRUCTURAL AXES

Work	Cls	Adaptation variable	Adaptation signal	Evidence base	Reported metric	Clin.
Chache *et al.* [8]	A	Coding rate	SNR, age of information	Analytical + bench testbed	Age of information	No
Taleb *et al.* [14]	A	Spreading factor	**Early Warning Score**, distance, PER	MIMIC recordings + analysis	PER, energy	**Yes**
González-Palacio *et al.* [9]	A	Tx power, data rate	ML-predicted SNR from environment	6-month field campaign	Energy, PDR, collisions	No
Kufakunesu *et al.* [11]	A	Spreading factor (randomised)	Interference, collision state	Network simulation	PDR, energy	No
Leonardi *et al.* [12]	A	Data rate	LSTM over link history	Network simulation	PDR, energy	No
Fernandes *et al.* [13]	A	Medium-access mode	Network load	Network simulation	Throughput, PDR	No
Ali *et al.* [15]	B	Transmission schedule	GMM traffic clustering	Network simulation	Collisions, delay, PSR	No
Dewa *et al.* [19]	B	Transmission order	**Medical urgency class**	Network simulation	Priority metric, throughput	**Sched.**
Ahammed & Manjaiah [20]	B	Transmission order	**Signal deviation, patient priority**	Simulation	QoS, energy, security	**Sched.**
Anand & Karthikeyan [21]	B	Task schedule, resources	Task priority, latency budget	Simulation	Latency, utilisation	Partial
Kang *et al.* [16]	B	Relay topology	Energy imbalance at cell edge	Simulation + testbed	Lifetime, energy	No
Mahalakshmi & Lalithamani [17]	B	Gateway placement	Coverage, connectivity, energy	OMNeT++ / FLoRa	Coverage, connectivity	No
Sarma Lanka & Kothandaraman [18]	B	Handover behaviour	Mobility state	Network simulation	Latency, reliability	No
Abdulmalek *et al.* [25]	C	None (fixed)	—	Hardware testbed (FiPy)	Reliability, latency, power	No
Bandyopadhyay *et al.* [23]	C	None (fixed)	—	Hardware testbed	Functional demonstration	No
Misran *et al.* [22]	C	None (fixed)	—	Hardware testbed	Range, functionality	No
Mdhaffar *et al.* [6]	C	None (fixed)	—	Hardware testbed	Feasibility	No
Jawad *et al.* [24]	C	None (fixed)	—	Hardware testbed	Range, reliability	No
Patel *et al.* [26]	C	None (fixed)	—	Testbed + hospital datasets	Fall detection, alerting	Alert only
Sayor & Rahman [27]	C	Bearer (Wi-Fi / LoRa)	Link availability	Hardware testbed (ESP32)	Connectivity, coverage	No
Nugraha *et al.* [28]	D	None (fixed)	—	Field trials, LOS / NLOS / crowded	Packet loss by site	No
Hadwen *et al.* [7]	D	GPS duty cycle	Energy budget	Hardware testbed	Energy, battery life	No
Pimentel *et al.* [29]	E	No radio	—	CapnoBase, MIMIC-II	Respiratory-rate error	n/a
Lin *et al.* [30]	E	No radio	—	CapnoBase, BIDMC (42 / 53 subj.)	Respiratory-rate error	n/a

The reported-metric column gives the quantity in which each work expresses its results, not a figure comparable across rows. The clinical-coupling column records whether a patient-derived variable reaches a communication decision: **Yes** where it selects a physical-layer parameter, **Sched.** where it reorders transmissions only, *Alert only* where a clinical inference raises an alarm without influencing the link, and *n/a* where the work contains no radio. PER denotes packet error rate, PDR packet delivery ratio, PSR packet success ratio, and LOS / NLOS line-of-sight and non-line-of-sight conditions.

band, and class D transmitting patient location in plaintext [7], [28].

A. Limitations of This Review

Four limitations bound everything claimed above. The corpus is 24 works on a DOI-verified, English-language protocol, and two LoRa healthcare papers were discarded solely because their journals do not deposit with Crossref, so the system-implementation class is slightly under-represented. Classes D and E carry two works each, enough to position them within the chain but not to survey them; nothing here is a claim about the current state of LoRa patient tracking or of photoplethysmographic estimation. The clinical-coupling counts are sensitive to their denominator, as Section IV states: one in 24 over the corpus, one in 13 over the works that adapt anything, and two of that subset [13], [16] were never attempting clinical coupling. And this review reports no experiments of its own, so every finding is a reading of published work rather than a measurement of what a new deployment would show.

VI. CONCLUSION

This review read LoRa-based remote health monitoring as one chain — sensing, physiological inference, link adaptation, network delivery — rather than as the five separate literatures it is usually surveyed in, across 24 DOI-verified works published between 2017 and 2026. Each layer is individually capable: physical-layer adaptation has moved from the signal-to-noise heuristic to learned controllers that predict link state directly, with reported energy savings approaching half at near-unity delivery ratio; network-level work has shown that scheduling, relaying and gateway placement all materially change what a dense deployment achieves; the deployed systems demonstrate that physiological telemetry over LoRa works; and the inference layer validates against annotated public recordings with stated subject counts.

The layers have not been joined. Clinical state reaches a physical-layer parameter decision in one of the 24 works, and a scheduling decision in two more; six of the seven end-to-end systems transmit on a configuration fixed at design time. The adaptation literature and the deployment literature report no metric in common, one being predominantly simulation and the other predominantly single-gateway hardware, so no work scores the chain from sensor waveform to cloud record end to end. One work in 24 addresses security, although every deployed system carries identifiable health data over a shared unlicensed band. And the duty-cycle ceiling that bounds how quickly an urgent event can be reported is documented in the background literature but measured in none of the surveyed systems.

TABLE III
THE FOUR OPEN CHALLENGES, THEIR EVIDENCE BASE, AND THE CLASSES EACH AFFECTS

Open challenge	Evidence in this corpus	Classes
No end-to-end evaluation protocol	Seven mutually non-composing metrics across the five classes; no work reports a sensor-to-cloud figure of merit	All five
Clinical state absent from link control	1 of 24 selects a PHY parameter from patient state [14]; 2 more reorder transmissions [19], [20]; 6 of 7 deployed systems are fixed-configuration	A, B, C
Duty cycle versus clinical latency	Airtime limits documented in [4], [5]; no corpus work reports urgent-event latency under an enforced duty cycle	A, B, C
Scale and security asymmetries	Network proposals at high density [15], [17], [19] against single-gateway deployments [22], [27]; 1 of 24 treats security [20]	B, C, D

Two steps follow directly. The first is a stated, evaluated policy mapping clinical urgency onto a LoRa configuration across all four parameters rather than spreading factor alone, taken on the patient-side node where classifier output and link estimate are both available. The second is an evaluation protocol reporting classification correctness and delivery latency as one figure of merit under an enforced duty cycle, so the cost of robustness can be weighed against the clinical value of the packet it is spent on. Until such a protocol exists, integrated remote-monitoring systems can be built, as several reviewed works show, but they cannot be compared.

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